G53MLE Machine Learning Past Year exam spring 2015 R4

Machine Learning (University of Nottingham Malaysia)

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, SPRING SEMESTER 2014-2015

MACHINE LEARNING (G53MLE)

Time allowed TWO hours

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer Question 1
AND
Section B: Answer TWO of the remaining three questions

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn your examination paper over until instructed to do so

ADDITIONAL MATERIAL:

- Exam paper
- Colour figure to be used for Question 3

INFORMATION FOR INVIGILATORS:

- Please ensure that Exam Paper and Answer Book are both submitted

Question 1: Foundations of Machine Learning [overall 25 marks]

(a) It is said that generalisation is the ultimate goal of machine learning. Explain what this term means, and name two general techniques (i.e. not specific to any particular machine learning method) that will improve generalisation.

(5 marks)

(b) In a performance maximisation scheme, (i) explain what a local maximum is. Give your answer both in words and using a sketch/drawing. Also explain in words (ii) why local maxima are problematic, and (iii) give a strategy to avoid local maxima.

(8 marks)

(c) Explain the difference between a linear binary classifier and a non-linear binary classifier. Draw two diagrams: one with a set of 2-dimensional data points of two classes for which a linear classifier would be a good solution, and one where a non-linear classifier would be the better choice.

(6 marks)

(d) A dataset consists of 500 elements. Using cross-validation, the sample error rate of hypothesis h1 is found to be 0.08 and that of hypothesis h2 is 0.11. Calculate the confidence intervals of the two errors and comment how that relates to the statistical difference between the results. z_N = 1.96 for a confidence level of 95%.

(6 marks)

SECTION B

Question 2: Artificial Neural Networks [overall 25 marks]

Below is a very simple data set encoding an XOR pattern.

INPUT 1	INPUT 2	OUTPUT
0	0	0
0	1	1
1	0	1
1	1	0

(a) Draw a diagram of an ANN's topology that can learn this pattern based on the given data, using a single hidden layer with three units. Name all relevant elements. Initialise all weights to 1.

(3 marks)

(b) For a given machine learning problem, explain the relation between the sample distribution and the true distribution of that problem.

(2 marks)

(c) For this problem, how can you sensibly increase the set of data? Explain the nature of this dataset in terms of how it represents the underlying distribution from which it's sampled.

(3 marks)

(d) Draw a single ANN unit ('neuron'), clearly indicating inputs, outputs, and clearly segregating linear combinations of signals from the non-linear components. Assign appropriate variable and function names and include these in your diagram

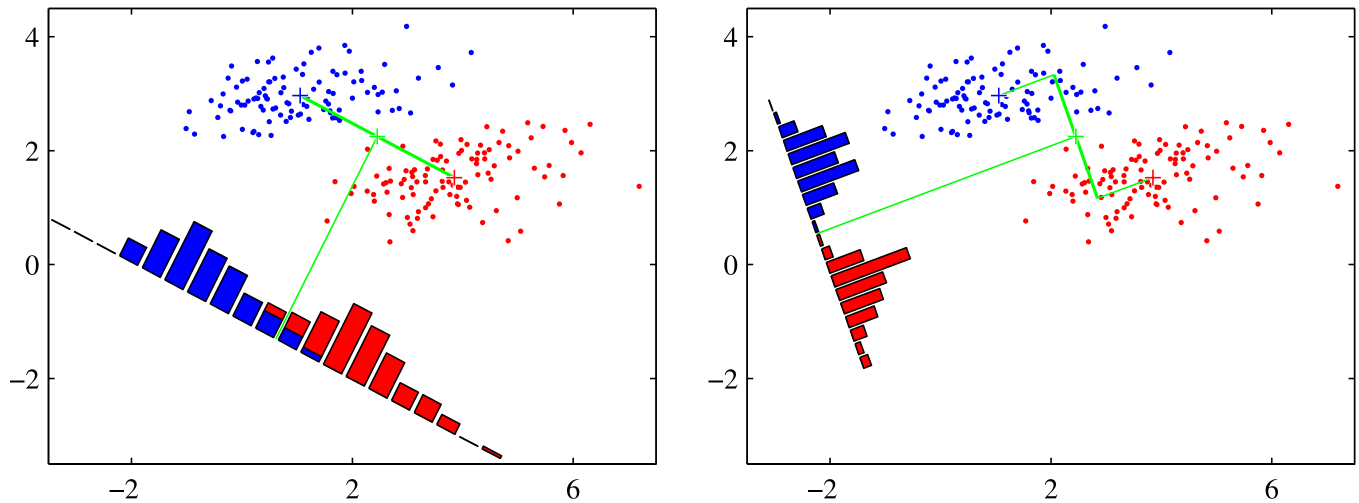
(5 marks)

(e) Assume the units have no non-linear activation function, and that you will use stochastic (online) gradient descent working through the data in sequence. Calculate back-propagation for the first two data points of Table 1 and the topology as described in Q2 (a) above. For both rounds, give (i) the errors for both input and output units, (ii) the error derivatives and (iii) the weight updates and (iv) new weights of each of the links, assuming a learning rate of $\eta = 0.1$.

(12 marks)

Question 3: Linear Classification and Regression [overall 25 marks]

Below is a picture of the result of Fisher's Linear Discriminant, which is used for Question 3 (a) and Question 3 (b). A colour version of this image is provided as an appendix on a separate piece of paper.



(a) Give the formula of the Fisher criterion J(w), and explain what each of the variables stands for.

(4 marks)

(b) Explain why Fisher's LDA can effectively be viewed as a dimensionality reduction algorithm. What is the dimensionality of the space Fisher's LDA projects data to?

(4 marks)

The table below defines a training set relating house prices (*y* values) to the square footage of each house (*x* values).

Square footage	House price
1	3.4
2	5.8
3	13.2
4	19.1
5	29
6	39.4
7	52.8
8	70.2
9	86.6
10	108

(c) After training a first-order polynomial model $h(x) = \theta_1 x + \theta_0$ and a second-order polynomial model $h(x) = \theta_2 x^2 + \theta_1 x + \theta_0$ the optimal parameters of the second order

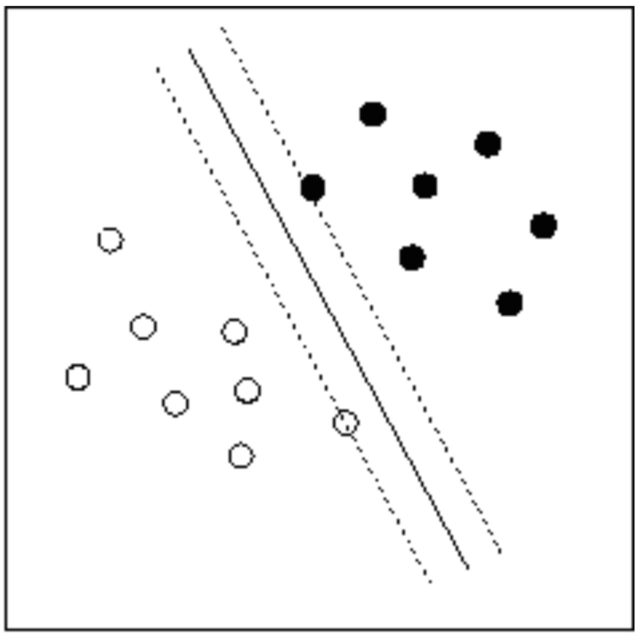
$$h(x) = \{\theta_2 = 1, \theta_1 = 0.4, \theta_0 = 2\}$$

model are found to be and those for the first-order model are found to be $h(x) = \{\theta_1 = 10, \theta_0 = -10\}$. Give the formulation and numeric value of the quadratic error e for both models on the training set for both models given these parameters, and explain what fundamentally causes the difference.

(12 marks)

(d) Below is a set of data with a sub-optimal separating hyperplane. Explain why this is sub-optimal, and how SVMs solve this problem. What is this aspect of SVMs called?

(5 marks)



Question 4: Data sets and data mining [overall 25 marks]

Below is a set of data, which is used for Question 4 (a), Question 4 (b), and Question 4 (c).

$$A = \begin{bmatrix} 2 & 4 & 8 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 & 8 & 1 \\ 2 & 4 & 8 & 2 & 4 & 3 \\ 1 & 1 & 1 & 1 & 2 & 2 \end{bmatrix}$$

(a) If we assume row-wise instances, what is the dimensionality of this dataset? How many data points does the dataset contain?

(2 marks)

(b) Explain the purpose of pre-processing in Data Mining, giving at least four purposes. If the data above were from a data mining problem, what pre-processing should you readily perform on the data?

(6 marks)

(c) Explain why matrix A presents an ill-posed problem. Give an explanation of one technique to remedy ill-posed problems.

(5 marks)

(d) Describe the difference between Data Mining and Machine Learning

(5 marks)

(e) For the set of data X presented below, find the principal components. Clearly indicate which is the first PC, and which the second.

$$X = \begin{bmatrix} -4 & 4 \\ -3 & 0 \end{bmatrix}$$

(7 marks)

 End